JEMBY DIGITAL TWIN: Concept of Operations (CONOPS)

Interactive Electrical Digital Twin, Multi-Scale Graph Visualizer, & AI Analysis Engine

1. Executive Summary & Vision

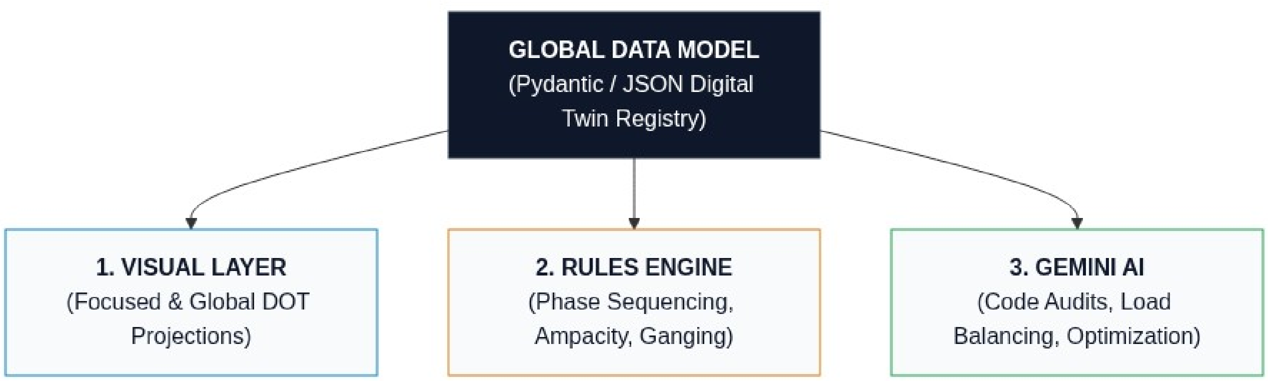
The **JEMBY Digital Twin** is an intelligent, physics-aware modeling platform for electrical power distribution systems. Unlike traditional static CAD software or abstract single-line diagram (SLD) tools, JEMBY treats electrical infrastructure as a **strongly-typed, relational object graph (Digital Twin)**.

Every component in a facility—from high-voltage utility feeds, transformers, switchboards, and distribution panelboards, down to embedded circuit breakers, branch conductors, and end-use loads—is instantiated as an autonomous object with defined electrical behaviors, physical containment hierarchies, and regulatory constraints.

This structured representation enables a three-pillar operational capability: 1. **Multi-Scale Visual Projections**: Dynamically rendering both high-level system topology (Global SLD) and high-fidelity physical equipment views (e.g., dual-column Panel Schedules with port-level wiring terminals) using browser-compiled Graphviz DOT. 2. **Deterministic Domain Rules Engine**: Automatically enforcing electrical physics (phase sequencing, ganged multi-pole trip mechanics, ampacity sizing, continuous load limits). 3. **Gemini AI Semantic & Analytical Engine**: Providing AI-driven NEC code compliance auditing, phase load balancing optimization, selective coordination validation, and natural-language design automation.

2. System Architecture: The Digital Twin Triad

The JEMBY architecture decouples data integrity from presentation and intelligence across three tightly integrated layers:



2.1 The Global Object Registry (Single Source of Truth)

Every electrical asset exists as a uniquely identifiable instance in a unified project graph. The model stores:

- **Intrinsic Attributes**: Electrical ratings (voltage, bus ampacity, kVA, AIC/SCCR, enclosure NEMA rating).
- **Containment Hierarchies**: Parent-child relationships (e.g., a Panelboard contains Slots; Slots contain Breakers).
- **Topology & Connectivity**: Explicit port-to-port connections via Conductor objects (e.g., `MDP:ckt_17_19` → `Feeder_10AWG` → `HVAC_Unit:L1_L2`).

2.2 Dual-Mode Visual Projections

The visual engine generates Graphviz DOT representations on demand from the underlying JSON state:

- **Global System SLD**: Macro-level overview showing power distribution from utility/generator down through switchboards, transformers, and subpanels.
- **Focused 1st-Degree Physical View (Drill-Down)**: Micro-level view placing a single selected equipment item at center stage, rendered with physical realism (e.g., dual-column panel schedule table), alongside its direct upstream sources and downstream branch loads.

2.3 Graphviz DOT HTML Table & Port Mapping

Visual realism and interactivity are achieved using Graphviz HTML-like label tables:

- Every circuit position is an explicit cell.
- Multi-pole ganged breakers span multiple rows or link with visual handle ties.
- Outgoing conductor edges bind strictly to breaker port terminals, preserving precise physical wiring origins.

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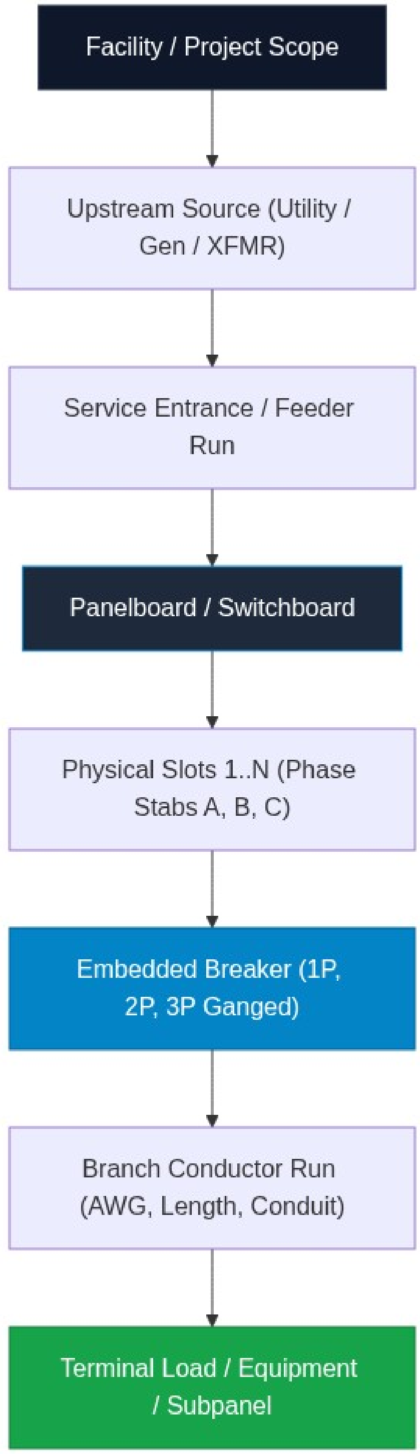
3. Physical Entity Hierarchy & Containment Model



3.1 Panelboard Object (`Panel`)

- **Identifiers**: `panel_id`, `tag` (e.g., `MDP-1`), `location`, `name`.
- **System Electrical Parameters**:
- Phase configuration: `1PH_3W` (120/240V) or `3PH_4W` (208Y/120V, 480Y/277V).
- Bus Ampacity: `100A`, `200A`, `225A`, `400A`, `800A`, etc.
- Mains Type: `MCB` (Main Circuit Breaker) or `MLO` (Main Lugs Only).

- Short-Circuit Current Rating (SCCR): e.g., `22kA`, `65kA`.
- Total Physical Spaces / Slots: `12`, `24`, `30`, `42`, `84`.

3.2 Embedded Breaker Object (`Breaker`)

Breakers live inside panel slots as first-class embedded objects:

- **Occupied Slots**: List of slot numbers (e.g., `[1]` for 1-pole; `[17, 19]` for 2-pole; `[1, 3, 5]` for 3-pole).
- **Poles & Ganging**:
- Number of poles: `1`, `2`, `3`.
- Common Internal Trip mechanism & physical handle tie linkage.
- **Ratings**:
- Trip Ampacity: `15A`, `20A`, `30A`, `50A`, `100A`, etc.
- Frame Size: `100A`, `225A`.
- Interrupting Capacity (AIC): `10kA`, `22kA`, `65kA`.
- Protection Type: Standard Thermal-Magnetic, GFCI, AFCI, Dual-Function (CAFCI/GFCI).
- **Operational State**: `ON`, `OFF`, `TRIPPED`, `SPARE`, `SPACE`.

3.3 Conductor / Branch Circuit Run (`Cable` / `Feeder`)

- **Conductor Specs**: Size (`14 AWG` to `500 kcmil`), Material (`Cu` / `Al`), Insulation (`THHN`, `XHHW-2`).
- **Installation**: Length in feet, Conduit type (`EMT`, `PVC`, `RMC`, `MC Cable`).
- **Calculated Values**: Conductor impedance, ampacity derating, calculated voltage drop (`V_d`).

3.4 Load / Downstream Asset (`Load`)

- Name, equipment category (Lighting, Receptacle, Motor, HVAC, EV Charger, Sub-panel).
- Operating Voltage, Connected kVA / kW, Power Factor (`PF`), Duty Cycle (Continuous vs Non-Continuous).

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4. Domain Intelligence & Physical Rules Engine

The Digital Twin automatically computes and enforces electrical principles:

4.1 Bus Stab Phase Sequencing

Panel interior stabs alternate phases down the column rows according to standard industry bus geometry:

Row Number	Left Slot	Right Slot	Single-Phase (120/240V)	Three-Phase (208Y/120V or 480Y/277V)
Row 1	Slot 1	Slot 2	**Phase A**	**Phase A**
Row 2	Slot 3	Slot 4	**Phase B**	**Phase B**
Row 3	Slot 5	Slot 6	**Phase A**	**Phase C**
Row 4	Slot 7	Slot 8	**Phase B**	**Phase A**
Row 5	Slot 9	Slot 10	**Phase A**	**Phase B**
Row 6	Slot 11	Slot 12	**Phase B**	**Phase C**

4.2 Multi-Pole Ganging Rules

• A **2-Pole Breaker** on the left column must occupy vertically adjacent odd slots (e.g., `17` & `19`), spanning **Phase A + Phase B** to deliver 240V (or 208V).

• A **3-Pole Breaker** must occupy three consecutive slots on the same side (e.g., `1, 3, 5`), spanning **Phase A + Phase B + Phase C**.

• A single load port (e.g., `port="ckt_17_19"`) is exposed for the ganged unit.

• Switching any pole to `OFF` or `TRIPPED` updates the entire ganged assembly synchronously.

4.3 Phase Load Balancing

• The system computes per-phase total volt-amperes (`VA_A`, `VA_B`, `VA_C`).

• Phase unbalance percentage is calculated continuously:

```
\% Unbalance = (\max(|VAA - VAavg|, |VAB - VAavg|, |VAC - VAavg|) / VAavg) x 100
```

• The model flags unbalance exceeding NEC/industry targets (`> 10\%`).

4.4 Sizing & Protection Constraints

• **Continuous Load Sizing (NEC 210.20)**: Overcurrent protection must be sized at `>= 125\%` of continuous load `+ 100\%` of non-continuous load.

• **Maximum Voltage Drop (NEC 210.19 Informational Note 4)**: Flags branch circuits where estimated drop exceeds `3\%` (or `5\%` total system drop).

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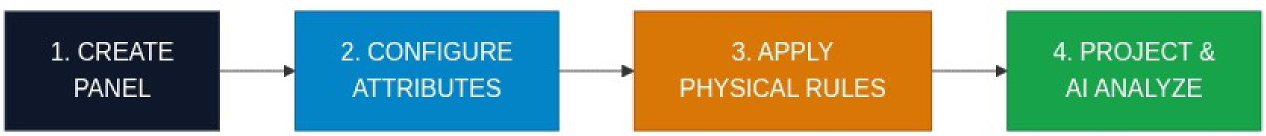
5. Panel Lifecycle & Interaction Flow



1. **Creation**: User instantiates a panel from the catalog (e.g., 24-space 200A 120/240V Panelboard). 2. **Attribute Configuration**:

• Set mains rating, enclosure, voltage system.

• Add/populate breakers in slots (1P, 2P, 3P, spare, space).

• Connect loads and branch conductors.

3. **Model & Rules Execution**:

• System validates slot availability and phase continuity.

• Auto-assigns phase legs to each breaker pole.

• Validates continuous load limits and bus capacity.

4. **Visual & AI Projection**:

• Generates Focused DOT HTML Schedule Table with clickable ports.

• Passes structured payload to Gemini for automated validation and optimization.

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6. Gemini AI Analytical Capabilities

By converting electrical systems into a structured digital twin, Gemini functions as an active engineering co-pilot:

1. **Automated NEC Code Audits**: Scans the entire project graph for violations (undersized conductors, overloaded breakers, improper neutral sizing, missing AFCI/GFCI protection in wet/living areas). 2. **Intelligent Load Balancing**: Recommends optimal breaker slot relocations to balance phase loads within `+/- 2\%`. 3. **Natural Language System Authoring**:

- *"Add a 48A Level 2 EVSE to Subpanel B. Find the best slot pair on underloaded phases, size the 2-pole breaker, and specify the conductor."*

4. **Selective Coordination & Fault Level Auditing**: Analyzes upstream-to-downstream trip curves to ensure faults isolate locally without taking down the main distribution board.

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7. Next Implementation Steps

1. **Panel & Breaker Data Schemas**: Define Pydantic models for `PanelConfig`, `SlotAssignment`, `BreakerObject`, and `Conductor` in `james_app/models.py`. 2. **Phase Determination Engine**: Build utility functions to compute phase assignments and ganged slot linkages. 3. **Focused Panel DOT Compiler**: Implement the HTML table schedule generator with dual-column layout and anchors. 4. **Interactive Schedule Editor**: Build an HTMX-driven dual-column panel schedule editor allowing direct breaker configuration and slot management.